CHAPTER 24 WAVE OPTICS

Key idea: light shows interference, diffraction and polarisation, so it is a transverse wave.

1 HUYGENS'S PRINCIPLE



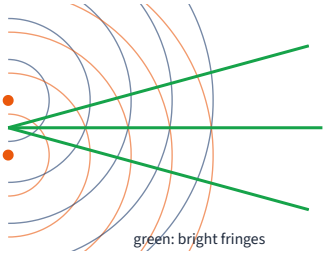
each point is a source of secondary wavelets; new wavefront = envelope

Every point on a wavefront is a source of secondary wavelets; the new wavefront is their **envelope**.

$$n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$$

On refraction f is unchanged; v and λ change.

2 YOUNG'S DOUBLE SLIT

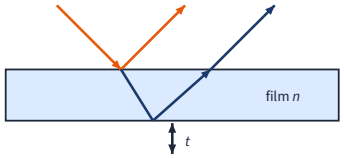


green: bright fringes

$$x = \frac{\lambda D}{a} \qquad a \sin\theta = n\lambda$$

- Bright: path difference $n\lambda$; dark: $(n + \frac{1}{2})\lambda$
- White light: white centre, coloured fringes (violet nearest)

3 THIN FILMS



path difference 2nt (+ λ/2 for one reflection phase change)

- π phase change on reflection at a **denser** medium
- Film in air (one change): bright $2nt = (m + \frac{1}{2})\lambda$
- Coating on glass (two changes): dark $2nt = (m + \frac{1}{2})\lambda$
- Minimum coating: $t = \dfrac{\lambda}{4n}$
- Air wedge: $x = \dfrac{\lambda}{2\theta}$, dark at contact

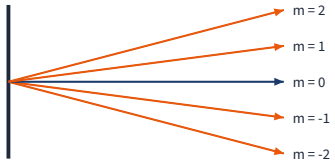
4 SINGLE-SLIT DIFFRACTION



$$a \sin\theta = m\lambda \qquad w = \frac{2\lambda D}{a}$$

Minima ($m = 1, 2, \ldots$). Central maximum twice as wide as the others.

5 DIFFRACTION GRATING

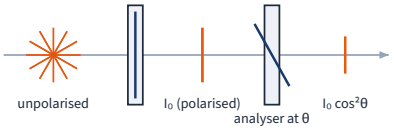


d sin θ = mλ: sharp, bright maxima

$$d \sin\theta = m\lambda \qquad d = \frac{1}{N}$$

- Highest order: largest integer below d/λ
- More lines: sharper, brighter maxima
- Total maxima $= 2m_{\max} + 1$

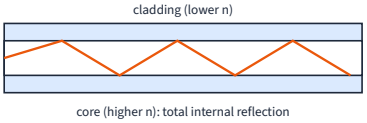
6 POLARISATION



$$I = I_0 \cos^2\theta \qquad \tan\theta_B = n$$

- Unpolarised through a polariser: $I_0/2$
- At θ_B: reflected and refracted rays at 90°
- Only **transverse** waves can be polarised

7 OPTICAL FIBRES



$$\sin\theta_c = \frac{n_2}{n_1} \qquad \mathrm{NA} = \sqrt{n_1^2 - n_2^2}$$

Total internal reflection: $n_{\text{core}} > n_{\text{cladding}}$, angle > θ_c. Modal dispersion spreads pulses; attenuation $= \dfrac{10}{L} \log \dfrac{P_{\text{in}}}{P_{\text{out}}}$ dB per unit length.

8 COMMON SECTION A TRAPS

- Mixing a (slit separation or width) with d (grating spacing).
- Forgetting the extra λ/2 from a reflection at a denser medium.
- Lines per mm → lines per m before $d = 1/N$.
- Using Malus's law on **unpolarised** light.
- Single slit: $a \sin\theta = m\lambda$ gives **minima**.

EXAM TIPS

- ✓State "coherent sources" and the path-difference condition.
- ✓For thin films, count the phase changes first.
- ✓Check $\sin\theta \leq 1$ for the highest grating order.
- ✓Fringe spacing ∝ λ: red fringes are wider than blue.